

Exact exponential periods, invariant constituent curvature, and collision homology

Split-Zero research programme

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1 Exact determinant of the polynomial–exponential period comparison

Fix the original monic cyclic polynomial

$$\chi(S) = S^q + \sum_{a=0}^{q-1} c_a S^a, \quad q \geq 1, \quad d = q + 1, \quad \Phi_t(S) = \frac{S^d}{d} + \sum_{a=0}^{q-1} \frac{c_a S^{a+1}}{a+1} - tS.$$

Fix $u \in \mathbb{C}^*$ and one argument of u . All the following comparisons retain this u , the original coordinate S , and the ordered remainder basis $1, S, \dots, S^{q-1}$. Set

$$D_{u,t}P = uP' + (\chi - t)P, \quad H_{u,t} = \mathbb{C}[S]/D_{u,t}\mathbb{C}[S].$$

This is a quotient of coefficient vector spaces. Let $\text{red}_{u,t}$ be its unique representative of degree less than q . Let $A(t)$ be the multiplication matrix of S on $\mathbb{C}[S]/(\chi - t)$ in the same ordered basis, so

$$A(t)e_b = e_{b+1} \quad (0 \leq b < q-1), \quad A(t)e_{q-1} = (-c_0 + t)e_0 - \sum_{a=1}^{q-1} c_a e_a. \quad (\text{XD1})$$

The empty sum convention in this display retains the case $q = 1$.

Use precisely the oriented rays and differences

$$\theta_j = \frac{\arg u + \pi}{d} + \frac{2\pi j}{d}, \quad \ell_j : S = re^{i\theta_j}, \quad 0 \leq r < \infty, \quad \Gamma_j = -\ell_0 + \ell_j \quad (1 \leq j \leq q). \quad (\text{XD2})$$

Write

$$\Pi_{jb}(u, c, t) = \int_{\Gamma_j} e^{\Phi_t(S)/u} S^b dS, \quad 1 \leq j \leq q, \quad 0 \leq b < q. \quad (\text{XD3})$$

1.1 Uniform estimates on the fixed unbounded contours

For a compact set $K \subset \mathbb{C}^{q+1}$ of the parameters $(c_0, \dots, c_{q-1}, t)$, put

$$M_K = \sum_{a=0}^{q-1} \frac{\sup_K |c_a|}{a+1} + \sup_K |t|, \quad R_K = \max\{1, 2dM_K\}, \quad \alpha = \frac{1}{2d|u|}. \quad (\text{XD4})$$

For every ray in (XD2) its leading term satisfies $S^d/(du) = -r^d/(d|u|)$. For $r \geq 1$ every other exponent has degree at most $d-1$, so

$$\text{Re} \frac{\Phi_t(re^{i\theta_j})}{u} \leq -\frac{r^d}{d|u|} + \frac{M_K r^{d-1}}{|u|} \leq -\alpha r^d \quad (r \geq R_K). \quad (\text{XD5})$$

Every parameter derivative of the integrand is its original exponential times a polynomial in S : a derivative in c_a contributes $S^{a+1}/((a+1)u)$ and a derivative in t contributes $-S/u$. For any fixed derivative order its absolute value on the tail is bounded by $Cr^Le^{-\alpha r^d}$, with C, L uniform on K . The required majorant is integrable, since substitution $x = \alpha r^d$ gives

$$\int_0^\infty r^L e^{-\alpha r^d} dr = \frac{1}{d} \alpha^{-(L+1)/d} \Gamma\left(\frac{L+1}{d}\right) < \infty. \quad (\text{XD6})$$

The remaining finite interval is uniformly bounded by continuity. Thus (XD3) converges absolutely and is entire in all the lower coefficients and t ; every parameter derivative may be taken under the integral. The same bound gives $e^{\Phi_t(S)/u} P(S) \rightarrow 0$ at the infinite endpoint of each ray, for every polynomial P . Consequently

$$\int_{\Gamma_j} e^{\Phi_t/u} D_{u,t} P dS = u \int_{\Gamma_j} d(e^{\Phi_t/u} P) = 0. \quad (\text{XD7})$$

Indeed the integral of that derivative on each outward ray is $-e^{\Phi_t(0)/u} P(0) = -P(0)$; the coefficients $-1, +1$ of the two rays cancel these values. These estimates handle the noncompact contours directly, including parameter values at which critical points coincide.

1.2 An exact trace comparison for twisted reduction

For $1 \leq m \leq q$ and $0 \leq b < q$, ordinary monic division gives unique polynomials $Q_{m,b}, R_{m,b}$ satisfying

$$S^{m+b} = (\chi - t)Q_{m,b} + R_{m,b}, \quad \deg R_{m,b} < q. \quad (\text{XD8})$$

If $m+b < q$, take $Q_{m,b} = 0$. Otherwise $\deg Q_{m,b} = m+b-q \leq q-1$, and the differential itself gives

$$\text{red}_{u,t}(S^{m+b}) = R_{m,b} - uQ'_{m,b}. \quad (\text{XD9})$$

Both terms on the right have degree below q . In every nonzero case,

$$\deg Q'_{m,b} \leq m+b-q-1 \leq b-1. \quad (\text{XD10})$$

Thus the coefficient of S^b in the derivative correction vanishes. Let $T_m(u, c, t)$ have column b equal to the coordinates of the left side of (XD9). The columns $R_{m,b}$ are exactly those of $A(t)^m$. Although (XD9) retains a possibly nonzero correction matrix, (XD10) proves the exact trace identity

$$\text{Tr } T_m(u, c, t) = \text{Tr } A(t)^m \quad (1 \leq m \leq q). \quad (\text{XD11})$$

Set $B_a = T_{a+1}/((a+1)u)$ for $0 \leq a < q$. Differentiation of (XD3), followed by (XD7)–(XD9), now proves

$$\partial_{c_a} \Pi = \Pi B_a, \quad \partial_t \Pi = -\Pi A(t)/u, \quad \text{Tr } B_a = \frac{\text{Tr } A(t)^{a+1}}{(a+1)u}. \quad (\text{XD12})$$

All the coefficient matrices here have polynomial entries in c, t for this fixed nonzero u .

1.3 The determinant, including every monomial factor

Define the scalar polynomial in the original coefficients

$$\mathcal{F}(c, t) = \text{Tr } \Phi_t(A(t)) = \frac{\text{Tr } A(t)^{q+1}}{q+1} + \sum_{a=0}^{q-1} \frac{c_a \text{Tr } A(t)^{a+1}}{a+1} - t \text{Tr } A(t). \quad (\text{XD13})$$

There is no choice or enumeration of roots in this definition. For any differentiable matrix A and integer $n \geq 1$, the product rule and cyclic invariance of trace give

$$d \operatorname{Tr} A^n = \sum_{r=0}^{n-1} \operatorname{Tr}(A^r (dA) A^{n-1-r}) = n \operatorname{Tr}(A^{n-1} dA). \quad (\text{XD14})$$

Because $A(t)$ acts on the actual quotient by $\chi - t$, it satisfies $\chi(A(t)) - tI_q = 0$. Applying (XD14) to (XD13) therefore gives

$$d\mathcal{F} = \sum_{a=0}^{q-1} \frac{\operatorname{Tr} A(t)^{a+1}}{a+1} dc_a - \operatorname{Tr} A(t) dt + \operatorname{Tr}((\chi(A(t)) - tI_q) dA(t)) = \sum_{a=0}^{q-1} \frac{\operatorname{Tr} A(t)^{a+1}}{a+1} dc_a - \operatorname{Tr} A(t) dt. \quad (\text{XD15})$$

At $c = t = 0$ the period matrix is

$$\Pi_{j,l-1}^{\text{mon}}(u) = a_l(u) e^{il\theta_0} (\zeta^{jl} - 1), \quad a_l(u) = \frac{(d|u|)^{l/d}}{d} \Gamma(l/d), \quad \zeta = e^{2\pi i/d}, \quad 1 \leq j, l \leq q. \quad (\text{XD16})$$

This follows by substituting $x = r^d/(d|u|)$ on each oriented ray in (XD3). In particular all $a_l(u)$ are positive real numbers. For $V_{jl} = \zeta^{jl} - 1$ one may include $j = 0$ in the Gram sum because its summand is zero. The complete root-of-unity sums give

$$(V^*V)_{lm} = \sum_{j=0}^{d-1} (\zeta^{-jl} - 1)(\zeta^{jm} - 1) = d\delta_{lm} + d. \quad (\text{XD17})$$

Here $1 \leq l, m < d$, so the two individual nonconstant sums vanish, and the first sum is $d\delta_{lm}$. The all-ones rank-one matrix has eigenvalue q on the all-ones vector and zero on its $(q-1)$ -dimensional perpendicular complement. It follows that

$$\det(V^*V) = d^q(1+q) = d^d > 0, \quad C_q(u) := |\det \Pi^{\text{mon}}(u)|^2 = d^d \prod_{l=1}^q \left[\frac{(d|u|)^{l/d}}{d} \Gamma(l/d) \right]^2. \quad (\text{XD18})$$

The complex determinant itself retains the oriented phase: $\det \Pi^{\text{mon}} = \det V \prod_{l=1}^q a_l(u) e^{il\theta_0}$.

We claim the exact global formula

$$\boxed{\det \Pi(u, c, t) = \det \Pi^{\text{mon}}(u) \exp(\mathcal{F}(c, t)/u)}. \quad (\text{XD19})$$

To prove it without assuming invertibility along the path, join $(0, 0)$ to (c, t) by (sc, st) , $0 \leq s \leq 1$. Equations (XD12) give $\Pi' = \Pi B(s)$, with continuous $B(s)$, and (XD15) gives $\operatorname{Tr} B(s) = \mathcal{F}'(sc, st)/u$. For every square matrix, including a singular one,

$$\frac{d}{ds} \det \Pi = \operatorname{Tr}(\operatorname{adj}(\Pi) \Pi B) = \det \Pi \operatorname{Tr} B. \quad (\text{XD20})$$

Consequently $\det \Pi \exp(-\mathcal{F}/u)$ has derivative zero. At the starting point A is nilpotent and $\mathcal{F}(0, 0) = 0$, so its value is $\det \Pi^{\text{mon}}(u)$. This proves (XD19). Equation (XD18) proves its nonvanishing for every c, t , including every repeated-root fibre. In particular the period map

$$H_{u,t} \longrightarrow \mathbb{C}^q, \quad [P] \longmapsto \Pi \operatorname{red}_{u,t}(P), \quad (\text{XD21})$$

is an isomorphism with inverse given by Π^{-1} followed by the specified degree-below- q section.

1.4 Exact determinant transport to the arithmetic representative

Retain the original positive Gram G_N on the cyclic coefficient space E , its determinant $V_N = \det G_N$, and the original maps $r_N : E \rightarrow \mathcal{B}^{\otimes k}$ and j_E on the admitted test-function domain with $j_E r_N = I_E$. Identify E with $H_{u,t}$ by their specified ordered remainder bases. Then the actual maps

$$r_N(E) \rightarrow \mathbb{C}^q, \quad x \mapsto \Pi j_E x, \quad \mathbb{C}^q \rightarrow r_N(E), \quad y \mapsto r_N \Pi^{-1} y \quad (\text{XD22})$$

are inverse: their composites are respectively $r_N j_E$ on $r_N(E)$ and $\Pi(j_E r_N) \Pi^{-1}$ on $\mathbb{C}^q$. On the full admitted domain the first observation has precisely $\ker j_E$ as its kernel; all original theta boundaries killed by j_E remain in that kernel.

With the explicitly fixed coordinate metric I_q on the target, put

$$H_\Pi = \Pi^* \Pi, \quad B_N^{\text{per}} = G_N^{-1} H_\Pi. \quad (\text{XD23})$$

These operators satisfy $G_N B_N^{\text{per}} = (B_N^{\text{per}})^* G_N = H_\Pi > 0$. Taking determinants in (XD19) gives the exact identities

$$\det H_\Pi = C_q(u) \exp\left(2 \operatorname{Re} \frac{\mathcal{F}(c, t)}{u}\right), \quad (\text{XD24})$$

$$\boxed{\log \det B_N^{\text{per}} = \log C_q(u) + 2 \operatorname{Re} \frac{\mathcal{F}(c, t)}{u} - \log V_N.} \quad (\text{XD25})$$

Every determinant in (XD24)–(XD25) is positive real; these logarithms therefore require no branch choice. At the same u, c, t for two admitted degrees $i \leq j$, subtraction proves

$$\boxed{\log \det B_j^{\text{per}} - \log \det B_i^{\text{per}} = \log(V_i/V_j).} \quad (\text{XD26})$$

Thus the common polynomial–exponential factor is now calculated exactly while the full original endpoint cost is retained by the displayed pair of positive comparison operators. No uniform upper estimate on V_i/V_j is inferred from this determinant identity.

2 SGA trace, exact constituent curvature, and the derived collision fibre

This calculation extends the constituent period map in the delivered *SGA Trace Period Control*, equations (54a)–(54c). Its period determinant is the same determinant already proved in (XD1)–(XD26), with every original coefficient, ray and phase retained. We calculate the full normal derivative, its kernel and its norm. The result applies to the actual invariant constituents, including repeated primary factors.

2.1 The retained period map and arithmetic source

Keep the original monic polynomial and its increasing-power frame:

$$\chi(S) = S^q + \sum_{a=0}^{q-1} c_a S^a, \quad E = \mathbb{C}[S]/(\chi), \quad A = M_S, \quad q \geq 1, \quad e_0 = [1], \quad \ell([P]) = [S^{q-1}] \operatorname{rem}_\chi P. \quad (\text{SC1})$$

Thus $\ell : E \rightarrow \mathbb{C}$ is a specified linear functional and $\mathcal{R} = e_0 \ell : E \rightarrow E$ has rank one. Fix $u \in \mathbb{C}^*$, one argument of u , and the actual potential and contours

$$\Phi_t(S) = \frac{S^{q+1}}{q+1} + \sum_{a=0}^{q-1} \frac{c_a S^{a+1}}{a+1} - tS, \quad \theta_j = \frac{\arg u + \pi + 2\pi j}{q+1}, \quad \Gamma_j = -\ell_0^{\text{ray}} + \ell_j^{\text{ray}}. \quad (\text{SC2})$$

Here ℓ_j^{ray} is the outward ray $S = re^{i\theta_j}$, $r \geq 0$; it is distinct from the coefficient functional ℓ . The period matrix is

$$\Pi_{jb}(t) = \int_{\Gamma_j} e^{\Phi_t(S)/u} S^b dS, \quad 1 \leq j \leq q, \quad 0 \leq b < q. \quad (\text{SC3})$$

The full convergence and determinant proof is (XD4)–(XD21). To state its precise retained inputs here, on a compact coefficient set K its tail is bounded by $r^L \exp(-r^{q+1}/(2(q+1)|u|))$ after $r \geq \max(1, 2(q+1)M_K)$, where $M_K = \sum_a \sup_K |c_a|/(a+1) + \sup_K |t|$. Differentiation is therefore valid in every coefficient direction. Integration by parts cancels the two origin terms, and reduction of S^q uses $[\chi - t] = 0$ with no derivative correction. It follows that

$$\Pi'(t) = -\frac{1}{u} \Pi(t)(A + t\mathcal{R}). \quad (\text{SC4})$$

The proven determinant is

$$\det \Pi(t) = \det \Pi_{\text{mon}}(u) \exp\left(\frac{\text{Tr } \Phi_t(A + t\mathcal{R})}{u}\right), \quad |\det \Pi_{\text{mon}}(u)|^2 = (2\pi|u|)^q > 0. \quad (\text{SC5})$$

The complex initial matrix is explicitly

$$(\Pi_{\text{mon}})_{j,b} = \frac{((q+1)|u|)^{(b+1)/(q+1)}}{q+1} \Gamma\left(\frac{b+1}{q+1}\right) e^{i(b+1)\theta_0} (e^{2\pi i j(b+1)/(q+1)} - 1). \quad (\text{SC6})$$

The product form and the full complex phase in (SC5)–(SC6) are reused from (XD18)–(XD19). The evaluated squared constant is the additional gamma-product evaluation in the delivered SGA note. Here is every factor in that evaluation. Set $d = q+1$. The matrix $V_{jl} = e^{2\pi i j l/d} - 1$ has $V^*V = d(I_q + J_q)$ by the root-of-unity sum in (XD17), so $|\det V|^2 = d^{q+1}$. Since $2 \sum_{l=1}^{d-1} l/d = q$, its column factors give

$$|\det \Pi_{\text{mon}}|^2 = d^{q+1} \frac{(d|u|)^q}{d^{2q}} \left(\prod_{l=1}^{d-1} \Gamma(l/d) \right)^2. \quad (\text{SC6a})$$

The gamma multiplication formula, DLMF (5.5.6), is

$$\Gamma(dz) = (2\pi)^{(1-d)/2} d^{dz-1/2} \prod_{a=0}^{d-1} \Gamma(z + a/d). \quad (\text{SC6b})$$

At $z = 1/d$, its left side and the final product factor are $\Gamma(1) = 1$, so the remaining product equals $(2\pi)^{q/2} d^{-1/2}$, which is DLMF (5.5.7). In (SC6a) the complete power of d is $(q+1) + q - 2q - 1 = 0$. This proves $|\det \Pi_{\text{mon}}|^2 = (2\pi|u|)^q$ without a frame change or an omitted phase. Thus every $\Pi(t)$ is invertible, including at collisions. The full reference is <https://dlmf.nist.gov/5.5.E7>.

For every original admitted source degree N , retain

$$K_N = B_N O_N^{-1} B_N^*, \quad G_N = K_N^{-1}, \quad R_N = O_N^{-1} B_N^* G_N, \quad r_N = \mathcal{V} R_N, \quad j_E r_N = I_E. \quad (\text{SC7})$$

The source identities and primitive have their original types $j_E D^{(k)} = A j_E$ and $D^{(k)} r_N - r_N A = d K_N^{\text{prim}}$. The exact parameter action on the admitted source domain is

$$D_t^{(k)} = D^{(k)} + t r_N \mathcal{R} j_E, \quad j_E D_t^{(k)} = (A + t\mathcal{R}) j_E, \quad D_t^{(k)} r_N - r_N (A + t\mathcal{R}) = d K_N^{\text{prim}}. \quad (\text{SC8})$$

Indeed insertion of $j_E r_N = I_E$ proves each equality by multiplication; the added terms cancel in the last difference. In particular the normal map calculated below has an explicit lift to this source, with the complete original primitive and kernel of j_E retained.

2.2 Every invariant constituent has a nonzero top coefficient

Let $I : F \hookrightarrow E$ be an actual injective linear map whose image is A -invariant, with $AI = IA_F$, and put $p = \dim F$. For the moment suppose $0 < p < q$. The image $I(F)$ is an ideal of E : invariance under A implies invariance under every polynomial in A , and every element of E is such a polynomial applied to e_0 . Its inverse image in $\mathbb{C}[S]$ is an ideal containing (χ) . It is generated by a unique monic divisor g of χ ; writing $\chi = gh$ gives $\deg h = p$ and $\deg g = q - p$. The exact map is

$$\mathbb{C}[S]/(h) \longrightarrow I(F), \quad [a]_h \longmapsto [ga]_\chi. \quad (\text{SC9})$$

It is onto by the definition of (g) , and is injective because $gh \mid ga$ in the polynomial domain implies $h \mid a$. Consequently $[g], [Sg], \dots, [S^{p-1}g]$ is an ordered basis of $I(F)$. Its last member is monic of degree $q - 1$, while every earlier member has smaller degree. Hence the restricted functional $L = \ell I : F \rightarrow \mathbb{C}$ is onto. Also $e_0 \notin I(F)$, since an ideal containing the unit is the whole algebra. We have proved

$$\begin{aligned} \mathcal{R}I &= e_0 L, \quad \text{rank}(\mathcal{R}I) = 1, \quad \ker(\mathcal{R}I) = \ker L, \quad \dim \ker L = p - 1, \\ \mathbb{C}e_0 \cap I(F) &= \{0\} \quad \text{inside } E. \end{aligned} \quad (\text{SC10})$$

These arguments never replace a repeated factor by its square-free part. In the basis (SC9), L is exactly the last coordinate; in the original chosen basis of F it remains the literal row ℓI .

2.3 The exact normal derivative at every parameter

Give the target $\mathbb{C}^q$ the period-coordinate Hermitian metric I_q specified in the delivered note. Define

$$\begin{aligned} Y(t) &= \Pi(t)I, \quad H(t) = Y(t)^* Y(t), \quad P(t) = Y(t)H(t)^{-1}Y(t)^*, \\ N(t) &= I_q - P(t), \quad z(t) = N(t)\Pi(t)e_0. \end{aligned} \quad (\text{SC11})$$

Every $Y(t)$ is injective, so $H(t) > 0$. Multiplication shows $P(t)^2 = P(t) = P(t)^*$ and $P(t)Y(t) = Y(t)$; thus $N(t)$ is the orthogonal projection onto the complement of the displayed period subspace. The vector $z(t)$ is nonzero for every t : otherwise $\Pi(t)e_0$ lies in $\Pi(t)I(F)$, and invertibility of $\Pi(t)$ would give $e_0 \in I(F)$, contradicting (SC10).

Using invariance under the original A , equation (SC4) yields

$$Y'(t) = -\frac{1}{u}Y(t)A_F - \frac{t}{u}\Pi(t)e_0L, \quad \boxed{N(t)Y'(t) = -\frac{t}{u}z(t)L.} \quad (\text{SC12})$$

This holds for every complex t , with the same fixed constituent F of the original A . We have not assumed that F is invariant under $A + t\mathcal{R}$. Formula (SC12) computes exactly its failure of invariance after passing through the original periods. For $t \neq 0$ the normal derivative has image $\mathbb{C}z(t)$, kernel $\ker L$ and rank one. At $t = 0$ it vanishes. Differentiating (SC4) once more gives

$$Y''(0) = \frac{1}{u^2}\Pi(0)A^2I - \frac{1}{u}\Pi(0)\mathcal{R}I, \quad N(0)Y''(0) = -\frac{1}{u}z(0)L. \quad (\text{SC13})$$

The first term has image inside $Y(0)F$ because $A^2I = IA_F^2$. The normal acceleration therefore has rank exactly one, image $\mathbb{C}z(0)$ and kernel $\ker L$, including all repeated-root cases.

The actual original source lift of its parameter defect is $r_N e_0 L : F \rightarrow r_N(E)$. The map $x \mapsto N(t)\Pi(t)j_E x$ sends it to $z(t)L$. Thus its source, coefficient and period forms have the commuting factorization

$$F \xrightarrow{L} \mathbb{C} \xrightarrow{a \mapsto ar_N e_0} r_N(E) \xrightarrow{j_E} E \xrightarrow{N(t)\Pi(t)} \mathbb{C}^q. \quad (\text{SC14})$$

At the original Split-Zero support, every linear arrow lifts by $x^\bullet \mapsto f(x)^\bullet$ and $\tau \mapsto \tau$. In particular $\ker L$ maps to the supported zero; external absence retains its distinct image τ . No new zero-adjunction is used.

2.4 Curvature is an exact rank-one expression

Write $\partial_t = \frac{1}{2}(\partial_x - i\partial_y)$ and $\partial_{\bar{t}} = \frac{1}{2}(\partial_x + i\partial_y)$ for $t = x + iy$. Since Y is holomorphic, $\partial_t H = Y^* Y'$, $\partial_{\bar{t}} H = Y'^* Y$ and $\partial_{\bar{t}} \partial_t H = Y'^* Y'$. Differentiating $H^{-1} H = I_p$ gives $\partial_{\bar{t}} H^{-1} = -H^{-1} Y'^* Y H^{-1}$. The determinant derivative and cyclic invariance of trace therefore give

$$\begin{aligned} \partial_{\bar{t}} \partial_t \log \det H &= \text{Tr}(H^{-1} Y'^* Y' - H^{-1} Y'^* Y H^{-1} Y^* Y') \\ &= \text{Tr}(H^{-1} Y'^* N Y'). \end{aligned} \quad (\text{SC15})$$

There is no logarithm-branch choice: $\det H > 0$. Insert the full normal map (SC12), use $N^* = N = N^2$, and retain the rank-one factors to obtain

$$\partial_{\bar{t}} \partial_t \log \det H(t) = |t|^2 c_F(t), \quad c_F(t) = \frac{\|z(t)\|_{L^q}^2}{|u|^2} L H(t)^{-1} L^* > 0. \quad (\text{SC16})$$

Strict positivity follows because $z(t) \neq 0$, $L \neq 0$ and $H(t)^{-1} > 0$. Thus this curvature vanishes exactly at $t = 0$ and vanishes there to precisely quadratic order; it is strictly positive at every $t \neq 0$. For the dual determinant line with metric $(\det H)^{-1}$ the convention $c_1 = -(i/(2\pi)) \partial \bar{\partial} \log h$ gives

$$c_1 = \frac{i}{2\pi} |t|^2 c_F(t) dt \wedge d\bar{t}. \quad (\text{SC17})$$

In these coordinates $i dt \wedge d\bar{t} = 2 dx \wedge dy$; this records the orientation and factor explicitly.

For completeness the exact formula supplies a convergent Taylor statement. Entries of Π have convergent complex power series; entries of H , P , N and c_F are real analytic because $\det H$ is everywhere positive. On any closed disk $|t| \leq \varepsilon$, continuity of the first real derivatives gives a finite bound $M_\varepsilon = \sup \|\nabla c_F\|$. Integrating the derivative along the real segment $s \mapsto st$ proves $|c_F(t) - c_F(0)| \leq M_\varepsilon |t|$. Consequently

$$\left| \partial_{\bar{t}} \partial_t \log \det H(t) - |t|^2 c_F(0) \right| \leq M_\varepsilon |t|^3. \quad (\text{SC18})$$

This estimate follows from the actual analytic functions in (SC11), and (SC16) supplies the stronger identity at all parameters.

One may check the projector's first derivatives directly: $\partial_t P = N Y' H^{-1} Y^*$ and $\partial_{\bar{t}} P = Y H^{-1} Y'^* N$. They vanish at zero by (SC12), explaining the retained stationary tangent space at that parameter without deleting its acceleration.

2.5 Both original theta and period metrics remain explicit

On F retain the positive original theta Gram $G_{N,F} = I^* G_N I$ and the positive comparison operator

$$B_{N,F}(t) = G_{N,F}^{-1} H(t), \quad \mathcal{O}_{N,F}(t) = \log \frac{\det H(t)}{\det G_{N,F}}. \quad (\text{SC19})$$

Here $G_{N,F}$ is fixed as t varies. Thus (SC15)–(SC18) hold unchanged with $\mathcal{O}_{N,F}$ in place of $\log \det H$. For real t at zero and $u > 0$, (SC12) gives $H'(0) = -(A_F^* H(0) + H(0) A_F)/u$, hence

$$-u \mathcal{O}'_{N,F}(0) - kp = 2 \text{Re Tr } A_F - kp. \quad (\text{SC20})$$

The original constant k and its generalized eigenspace multiplicities are retained. On the full positive-real-defect constituent this is the existing aggregate defect $L_{h,k}$.

The squared Hilbert–Schmidt norm of the normal acceleration in the original domain metric, and its one nonzero squared singular value, are both

$$a_{N,F} = \frac{\|z(0)\|_{I_q}^2}{|u|^2} L G_{N,F}^{-1} L^*. \quad (\text{SC21})$$

Indeed the adjoint of $zL : (F, G_{N,F}) \rightarrow (\mathbb{C}^q, I_q)$ is $G_{N,F}^{-1} L^* z^*$, so the product and its trace give (SC21). The unique nonzero eigenvalue equals its trace because the product has rank one. The curvature coefficient is instead the exact observer

$$c_F(t) = \frac{\|z(t)\|_{I_q}^2}{|u|^2} L B_{N,F}(t)^{-1} G_{N,F}^{-1} L^*. \quad (\text{SC22})$$

This follows from $H = G_{N,F} B_{N,F}$; it is the required map between the two metric measurements. It does not replace either metric by an unweighted coordinate norm.

Finally transport the original metric on the whole period target by $\tilde{G}_N = \Pi^{-*} G_N \Pi^{-1}$. Its projection onto YF is

$$P_N^\theta = Y G_{N,F}^{-1} Y^* \tilde{G}_N = \Pi I G_{N,F}^{-1} I^* G_N \Pi^{-1}. \quad (\text{SC23})$$

Direct multiplication proves its idempotence and its $\tilde{G}_N$ -self-adjointness. With $z_N^\theta = (I_q - P_N^\theta) \Pi e_0$, its squared target norm is the literal theta Schur complement

$$\|z_N^\theta\|_{\tilde{G}_N}^2 = e_0^* G_N e_0 - e_0^* G_N I G_{N,F}^{-1} I^* G_N e_0 > 0. \quad (\text{SC24})$$

The right side is positive because it is the norm of the nonzero G_N -orthogonal projection of e_0 away from $I(F)$. For this moving target metric $Y^* \tilde{G}_N Y = G_{N,F}$ is constant. Its differentiated metric terms are explicitly

$$\partial_t \tilde{G}_N = \Pi^{-*} G_N (A + t\mathcal{R}) \Pi^{-1} / u, \quad \partial_t \tilde{G}_N = \Pi^{-*} (A + t\mathcal{R})^* G_N \Pi^{-1} / \bar{u}. \quad (\text{SC25})$$

They cancel $Y^* \tilde{G}_N Y'$ and its conjugate derivative in the product rule. Thus the constant pulled-back theta Gram and the nonzero curvature (SC16) are connected by the exact metric congruence (SC23)–(SC25), with every term retained.

For $F = 0$, use the empty Gram determinant 1, empty trace 0 and the zero normal map. For $F = E$, $P = I_q$ and $N = 0$ identically, so (SC15) is zero: the full determinant logarithm in (SC5) is the real part of a holomorphic function plus its stated constant. The strictly positive coefficient in (SC16) belongs precisely to $0 < p < q$. This curvature and the original theta comparison are fully calculated observables; no uniform arithmetic upper bound is asserted by these finite-dimensional identities.

2.6 The exact derived base change at a collision

The trace and the positive-degree diagonal groups in the delivered SGA note have a further explicit specialization map. Retain the whole one-parameter algebra over $T = \mathbb{C}[t]$:

$$\mathcal{E} = T[S]/(\chi(S) - t) \simeq \mathbb{C}[S], \quad t \mapsto \chi(S), \quad M = \mathcal{E}/\chi' \mathcal{E}. \quad (\text{SC26})$$

The isomorphism sends $[P(t, S)]$ to $P(\chi(S), S)$; its inverse sends S to its original class. Monic division proves that $\mathcal{E}$ is free of rank q over T . Since χ' is a nonzero polynomial over $\mathbb{C}$, multiplication by χ' is injective on $\mathcal{E} \simeq \mathbb{C}[S]$. Consequently

$$0 \longrightarrow \mathcal{E} \xrightarrow{\chi'} \mathcal{E} \longrightarrow M \longrightarrow 0 \quad (\text{SC27})$$

is a free T -resolution of M . The diagonal resolution gives $HH_0(\mathcal{E}/T) = \mathcal{E}$, $HH_{2a+1}(\mathcal{E}/T) = M$ and $HH_{2a+2}(\mathcal{E}/T) = 0$ for $a \geq 0$. For clarity, that resolution is free over $\mathcal{B} = \mathcal{E} \otimes_T \mathcal{E}$ with alternating maps $X - Y$ and $\mathcal{C}_H = (H(X) - H(Y))/(X - Y)$, followed by the multiplication augmentation. Over $(T[Y]/H(Y))[X]$ both factors are monic and therefore non-zero-divisors; their product is $H(X)$ modulo $H(Y)$. Cancellation proves each alternating kernel equals the preceding image. Tensoring by the diagonal sets $X = Y$ and gives the maps $0, \chi', 0, \chi', \dots$, proving the stated groups.

Fix $t_0 \in \mathbb{C}$, put $\kappa = T/(t - t_0)$ and $E_{t_0} = \mathbb{C}[S]/(\chi - t_0)$. Tensoring (SC27) gives the complete kernel-and-cokernel sequence

$$\begin{aligned} 0 \longrightarrow \mathrm{Tor}_1^T(M, \kappa) &\longrightarrow E_{t_0} \xrightarrow{\chi'} E_{t_0} \\ &\longrightarrow M \otimes_T \kappa \longrightarrow 0. \end{aligned} \quad (\text{SC28})$$

It computes the actual specialized diagonal groups:

$$\begin{aligned} HH_{2a+1}(E_{t_0}/\mathbb{C}) &= M \otimes_T \kappa = E_{t_0}/(\chi'), \\ HH_{2a+2}(E_{t_0}/\mathbb{C}) &= \mathrm{Tor}_1^T(M, \kappa) = \mathrm{ann}_{E_{t_0}}(\chi'). \end{aligned} \quad a \geq 0. \quad (\text{SC29})$$

Every arrow is induced by the same polynomial multiplication, so these are equivariant maps for the original S -action.

There is also a representative-level description of the Tor map. Resolve κ by $[T \xrightarrow{t-t_0} T]$ to identify $\mathrm{Tor}_1^T(M, \kappa)$ with the kernel of $t - t_0$ on M . If $[v]_{\chi'}$ belongs to this kernel, write

$$(\chi - t_0)v = \chi'w. \quad [v]_{\chi'} \longmapsto [w]_{\chi-t_0}. \quad (\text{SC30})$$

Replacing v by $v + \chi'a$ replaces w by $w + (\chi - t_0)a$, so the map is well-defined. Its image is annihilated by χ' . If its image is zero, $w = (\chi - t_0)a$ and cancellation of the nonzero polynomial $\chi - t_0$ gives $v = \chi'a$, proving injectivity. Conversely an annihilated class $[w]_{\chi-t_0}$ satisfies $\chi'w = (\chi - t_0)v$ for a polynomial v , giving its preimage. These operations commute with multiplication by S .

At a root λ of multiplicity m in $\chi - t_0$, its local algebra is $\mathbb{C}[x]/(x^m)$, $x = S - \lambda$, and $\chi' = mx^{m-1}a(x)$ with the original invertible local factor a (this equality is in the local algebra: before reduction, the derivative of $x^m a(x)$ also contains the term $x^m a'(x)$). The odd group has local module $\mathbb{C}[x]/(x^{m-1})$ and the even group has local module $(x)/(x^m)$; both have dimension $m - 1$, with their full induced $S = \lambda + x$ action. The map $[b]_{x^{m-1}} \mapsto [xb]_{x^m}$ is an explicit module isomorphism between these local modules, using the stated local coordinate. For $m = 1$ both groups vanish. Thus the zero even group of the family and the nonzero even group of a collision fibre are related by the exact Tor term (SC28), not by omitting specialization data. The coefficient maps in the residue construction commute with base change; this calculation retains the additional derived term required when their kernels are specialized. Lifting every arrow at its original Split-Zero support keeps its receiving supported zero and external absence distinct as in (SC14).

2.7 The exact neighboring-minor identity and its theta factors

The ideal basis in (SC9) supplies three actual polynomial inclusions. Write their increasing-power coefficient matrices as

$$J_g = [g, Sg, \dots, S^{p-1}g], \quad J_- = [g, Sg, \dots, S^{p-2}g], \quad J_+ = [J_g, e_0]. \quad (\text{SC31})$$

For $p = 1$, J_- has zero columns. All three are injective on their stated coefficient spaces; the first two assertions follow from their distinct monic leading degrees, and the last from $e_0 \notin I(F)$. Let

$C = (I^*I)^{-1}I^*J_g$. Since the columns of J_g lie in $\text{im } I$, this satisfies $IC = J_g$; it is invertible because both matrices are bases of that same subspace. The exact frame equations are

$$H_g(t) := (\Pi J_g)^* \Pi J_g = C^* H(t) C, \quad LC = (0, \dots, 0, 1), \quad h_g(t) = |\det C|^2 \det H(t), \quad (\text{SC32})$$

where $h_J(t) = \det((\Pi J)^* \Pi J)$ for each of the three inclusions. The scalar $|\det C|^2$ has been retained; it is independent of t , so it has zero logarithmic derivatives.

The Gram of ΠJ_+ has block form

$$\begin{pmatrix} H_g & (\Pi J_g)^* \Pi e_0 \\ (\Pi e_0)^* \Pi J_g & \|\Pi e_0\|^2 \end{pmatrix}. \quad (\text{SC33})$$

Eliminate its off-diagonal blocks using triangular matrices of determinant one. Its lower-right Schur complement is the squared norm of the orthogonal projection of Πe_0 away from $\text{im } \Pi J_g$, which is $\|z(t)\|^2$ by (SC11). Thus $h_+ = h_g \|z\|^2$. The last diagonal entry of H_g^{-1} , computed by the adjugate formula, is its upper-left principal minor divided by h_g , namely h_-/h_g . By (SC32) that entry is exactly $LH^{-1}L^*$. Formula (SC16) becomes

$$\partial_t \partial_t \log h_g(t) = \frac{|t|^2}{|u|^2} \frac{h_+(t)h_-(t)}{h_g(t)^2}. \quad (\text{SC34})$$

This identity retains all three original polynomial maps. It is the precise neighboring-minor equation joining the period calculation to the source-volume and Toda calculations; it does not identify differently ordered or differently weighted families.

For each same inclusion $J = J_-, J_g, J_+$, define its original theta volume and its positive comparison determinant by

$$v_{N,J} = \det(J^* G_N J), \quad b_{N,J}(t) = \det((J^* G_N J)^{-1} J^* \Pi(t)^* \Pi(t) J). \quad (\text{SC35})$$

Injectivity and positivity show that both numbers are positive. For the empty inclusion both are 1. Taking the determinant of the literal product gives $h_J = v_{N,J} b_{N,J}$, and therefore

$$c_F(t) = \frac{1}{|u|^2} \frac{v_{N,J_+} v_{N,J_-}}{v_{N,J_g}^2} \frac{b_{N,J_+}(t) b_{N,J_-}(t)}{b_{N,J_g}(t)^2}. \quad (\text{SC36})$$

Every original theta mass and coefficient inclusion remains in the v factors; every period comparison remains in the b factors. This proves the metric dictionary required to use (SC34) within that original source programme.

For $p = q - 1$, the divisor g is monic linear. The square matrix $J_+ = [g, Sg, \dots, S^{q-2}g, e_0]$ has determinant $(-1)^{q-1}$: expand along its last column, and the remaining minor on rows $1, \dots, q - 1$ is upper triangular with every diagonal entry 1. Thus, with that sign retained,

$$\det(\Pi J_+) = (-1)^{q-1} \det \Pi, \quad h_+(t) = (2\pi|u|)^q \exp\left(2 \operatorname{Re} \frac{\operatorname{Tr} \Phi_t(A + t\mathcal{R})}{u}\right). \quad (\text{SC37})$$

The top minor in (SC34) is therefore the explicitly evaluated full determinant in this codimension-one case. The remaining two minors and their exact theta factors are still those in (SC31)–(SC36).